

Adolescent Alcohol Use & BMI Inference

CYCLE 3: CONFIDENCE INTERVAL & TWO-SAMPLE T-TEST

GROUP 8 | FINAL REPORT

STUDY CONTEXT

RESEARCH QUESTION

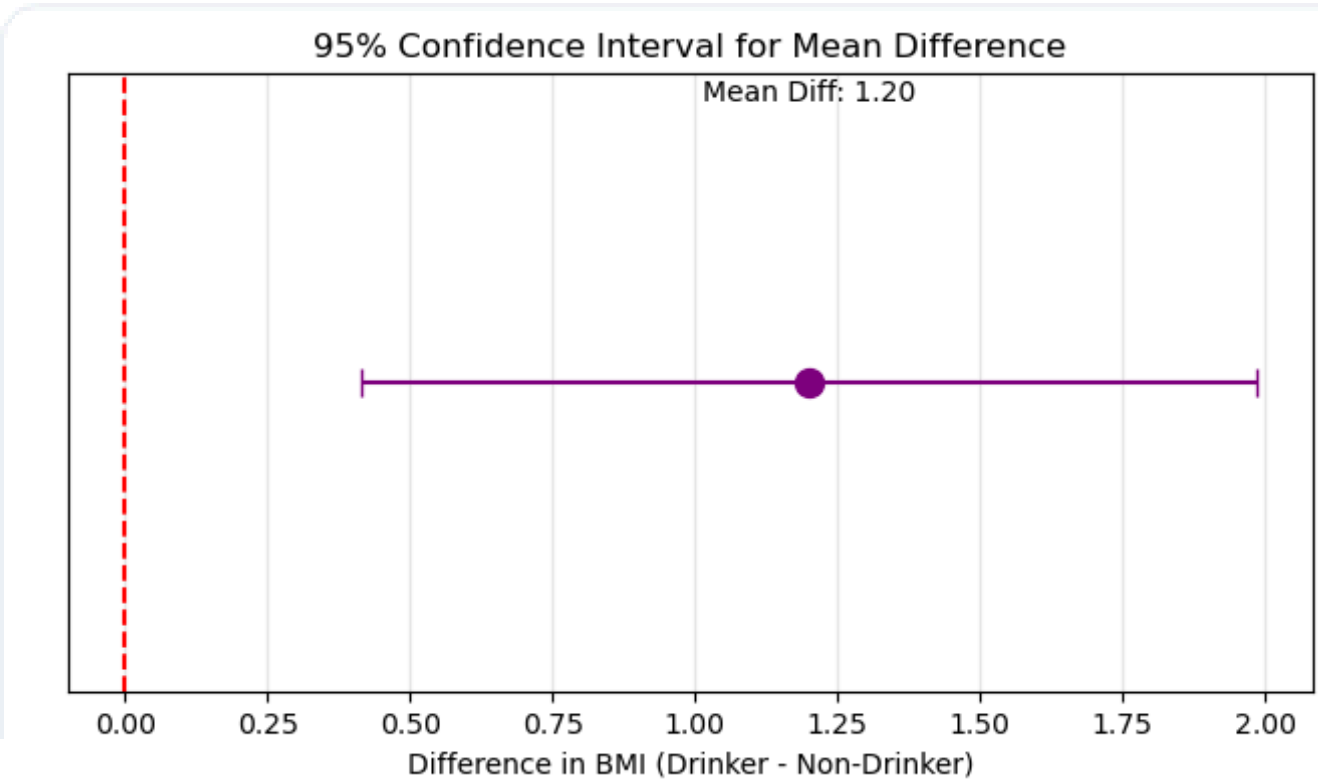
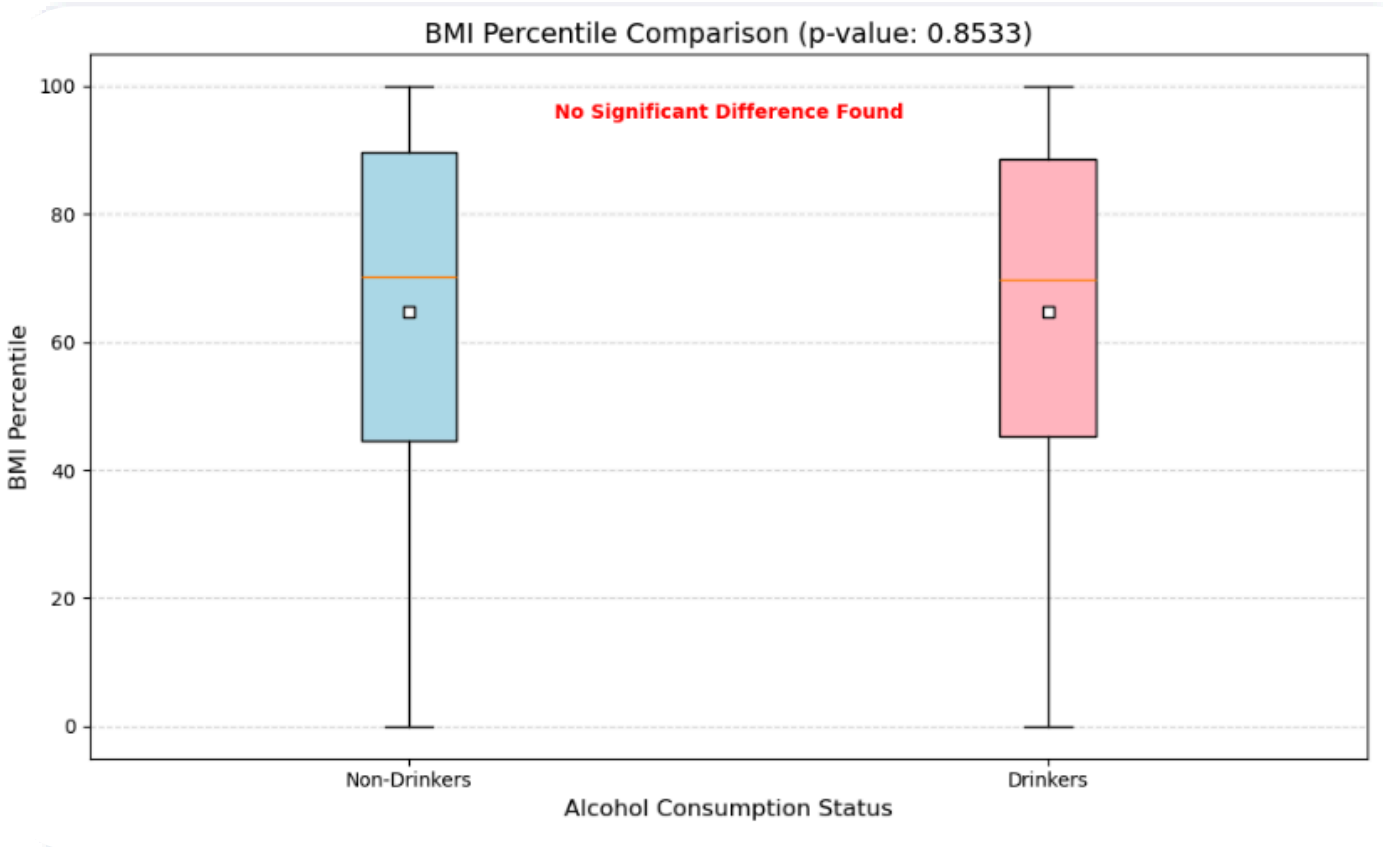
Does current alcohol consumption significantly impact BMI percentiles in US high school students?

SAMPLE PROFILE (YRBS 2007)

Group	n-Size	Mean BMI
Non-Drinkers (0)	6,449	64.71
Drinkers (1)	5,394	64.80

Observed Diff +0.0934

Note: BMI percentiles are based on age and sex. Analysis excludes missing values (N = 11,843).



INFERENCE RESULTS

P-VALUE (SIGNIFICANCE)

0.8533

Threshold $\alpha = 0.05$

95% CONF. INTERVAL

[-0.90, 1.08]

t-Statistic: 0.1849

Test: Welch Two-Sample t-Test

VERDICT: FAIL TO REJECT H_0

No statistically significant difference found. The 95% CI contains zero, and the high p-value suggests the 0.09 difference is likely due to sampling noise.